Organic Chemistry and Biomolecules

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Organic Chemistry and Biomolecules (A13123)

Short Title:	Organic Chem & Biomolecules		
Faculty	Science and Computing		
Department	Science		
Cluster	Chemistry		
Author	EMCNEELA		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module covers the main categories of organic compounds and important classes of biological molecules. Characterisation of synthetic organic molecules and analysis of biomolecules, including structure, nomenclature and stereochemistry, is examined. The laboratory element will include the techniques of synthesis, purification and characterisation.

Programmes

`CHEM-0006`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	2 / 3 / M
`CHEM-0006`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	2 / 3 / M
`CHEM-0006`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	2 / 3 / M
`CHEM-0006`	BSc in Pharmaceutical Science (`WD_SPHSC_D`)	2 / 3 / M

Indicative Content

- Stereochemistry, stereoisomerism and the nomenclature of stereoisomers
- Structure, properties and stereochemistry of biomolecules: carbohydrates, lipids, proteins
- An introduction to metabolism of important biomolecules within the body
- Important chemical and biochemical reactions, including substitution, elimination, oxidation and reduction
- Techniques for the purification of organic compounds
- Determination of product purity and identification using melting points, chromatography and spectroscopy
- Basic synthetic techniques and functional group tests
- Analysis and characterisation of biomolecules

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the stereochemical characteristics of organic molecules.
2. Describe the chemical reactions and physical properties of a range of organic compounds, classified according to their reaction type.
3. Sketch the structures of important biomolecules and recognize key functional groups within these molecules.
4. Compare and contrast fundamental properties of a range of biomolecules.
5. Apply the correct procedures for synthesis, purification and characterization of organic molecules.
6. Demonstrate adherence to safe laboratory procedures, including the correct disposal of laboratory waste.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Laboratory-based practicals.
- Self-directed study.
- Video tutorials and online animations.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	5,6
In-Class Assessment	30%	1,2,3,4
Tutorial/Problem Sheets	20%	1,2,3,4

Assessment Criteria

<40%: No understanding of the basic principles of organic chemistry and biomolecules. Inability to carry out associated laboratory work to a satisfactory standard.

40%-49%: Will be able to show a basic knowledge of organic chemistry and biomolecules. Demonstrate the ability to carry out associated practical work to a fair standard.

50%-59%: As well as the above, the student will be able to demonstrate an understanding of some of the complexities of organic chemistry and biomolecules and predict some trends in reactivity and behaviour. In addition, the learner will show a consistently good ability in associated practical work.

60%-69%: In addition, the student will demonstrate more detailed knowledge of organic chemistry and biomolecules and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material. Will demonstrate very good ability in most laboratory experiments.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of organic chemistry and biomolecules, predict general trends in behaviour, including exceptional behaviour and show the ability to evaluate and relate topics. Will demonstrate excellent laboratory skills and a consistently high ability in the analysis of results.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- "Biochemistry Free and Easy." http://biochem.science.oregonstate.edu/content/biochemistry-free-and-easy
- Bruice, P.Y. _Organic Chemistry_ . 5th Edition. USA: Pearson Education, 2007.
- Voet, D. _Fundamentals of Biochemistry_. Hoboken, N.J.: Wiley, 2006.

Supplementary Material

- McMurray, J.E. _Organic Chemistry_. 6th Edition. California: Brooks Cole, 2003.

Requested Resources

- Room Type: Chemistry Lab